

★ WEEK 2 · LECTURE

Financial Data Structure

Bars, the CUSUM filter — and why the clock is a bad ruler

Expanded edition · 38 slides · with three hands-on labs

Before any model, we must decide how to chop the market into rows of data.
We'll use party photos and smoke alarms to make every idea obvious.

Where We Are Going Today

Three simple questions — plus Week 1's rule: garbage data in, garbage model out

1

How should we chop the market into rows?

Party photos on a timer vs. photos when something happens.

2

How do we focus on moments that matter?

The CUSUM filter — a smoke alarm for prices.

3

What dirty-data traps fake great results?

Survivorship, look-ahead, and outliers — the usual suspects.

By the end you will build dollar bars and a CUSUM alarm in your browser.

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PART 1 · SLICING THE MARKET

How to Slice the Market

Bars: the rows of our data table

The Day the Market Fell 9% in Minutes

6 May 2010 — the 'Flash Crash' that exposed fragile, clock-based data

"At 2:42 pm the Dow dropped about 1,000 points, then bounced back minutes later. A daily bar barely noticed."

What happened

- One big automated sell order hit a thin market.
- Fast algorithms amplified it within seconds.
- Nearly \$1 trillion vanished, then mostly returned in ~30 minutes.

The lesson for us

- Market action is wildly uneven in time.
- A '1-day bar' treats a frantic minute and a sleepy hour as equals.
- We need to sample by activity, not by the clock.

The Flash Crash in Four Numbers

One afternoon that every data-sampling argument comes back to

~1,000

Dow points lost in minutes — about 9% of the index

36 min

From collapse to near-full recovery

~\$1 T

Market value that vanished and (mostly) came back

1 bar

What a daily chart shows — one long wick, story erased

A whole crisis can live and die inside a single time bar — activity is the real clock.

First — What Even Is a 'Bar'?

A bar is just one row in our data spreadsheet

A 'bar' summarizes a chunk of trading into one row: its Open, High, Low and Close price — think of one candle on a price chart.

The only question

- When do we 'close' one bar and start the next?
- Every N minutes? Every N trades? Every N dollars?
- That single choice shapes all our data.

Why beginners should care

- Most people blindly use daily (time) bars.
- We'll see they have hidden statistical problems.
- A smarter choice makes every later model better.

Party Photos: Timer vs. Action

Time bars = photos on a timer. Activity bars = photos when something happens.

Two Ways to Take Photos at a Party

A · Snap every 10 minutes (clock)

You take a photo on a fixed timer, no matter what is happening.

During the boring wait: 20 identical photos of an empty room.

During the surprise cake moment: maybe ZERO photos — you missed it.

This is a TIME bar. The clock decides, not the action.



B · Snap when something happens

You take a photo whenever real activity occurs.

Boring wait → few photos. Good.

Exciting cake moment → many photos.
You captured what mattered.

This is an ACTIVITY bar (tick / volume / dollar). The action decides.

Markets are exactly like this party.

Four Ways to Build a Bar

Tick / volume / dollar bars sample by activity — giving cleaner, more bell-curve-like returns.

Four Ways to Slice the Market into "Bars"

Time Bars

One bar per fixed clock interval (e.g. 1 day).

- + Simplest, most common
- Markets are not uniform in time: quiet nights, frantic opens
- Returns are far from i.i.d. (autocorrelated, heteroskedastic)

Tick Bars

One bar per N trades (transactions).

- + Sampled by activity, not the clock
- + Returns closer to a normal distribution
- Order fragmentation distorts trade counts

Volume Bars

One bar per N shares traded.

- + Tracks how much actually changed hands
- + Even better statistical properties than ticks
- Price level ignored: 10 vs 1000 dollar shares

Dollar Bars (best)

One bar per fixed dollar value traded.

- + Robust to price level and stock splits
- + Most stable bar count over the long run
- + López de Prado's default choice

Four Bar Types at a Glance

Same market, four rulers — keep this table next to the picture

Bar type	Closes when...	Strength	Weakness
Time	N minutes / 1 day passes	Simple, everywhere	Ignores activity; ugly stats
Tick	N trades have printed	Tracks trading pace	One 1-share trade = one 10,000-share trade
Volume	N shares have traded	Weighs trade size	Splits / price level distort it
Dollar	\$N of value has traded	Most stable unit over time	Needs a threshold choice

The professional default is the dollar bar — value traded is the most comparable unit.

Why 'Time' Is a Bad Ruler

First, the one thing we always measure: the return r — how much the price changed

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

The problem with time bars

- Markets digest news at an uneven pace, not on a clock.
- Time-bar returns swing between calm and wild (changing variance).
- Their tails are fatter than the bell curve models assume.

The fix: sample by activity

- A dollar bar closes each time a fixed dollar amount has traded.
- Busy times → many bars; quiet times → few bars.
- Returns come out much closer to a clean bell curve.

Log vs. Simple Returns — Why the log?

Both measure change; the log version is the one quants add up

Simple return: $(P1 - P0) / P0$. Log return: $\ln(P1 / P0)$. For small moves they are almost identical — so why bother with logs?

Three gifts of the log

- They ADD across time: a week's return = the sum of five daily logs.
- They are symmetric: +10% then -10% nets to exactly zero.
- They make the math of volatility and CUSUM clean.

When simple is fine

- Talking to clients: 'the fund made 12%' is a simple return.
- Portfolio weights combine with simple returns.
- Rule: model in logs, report in simple.

In this course, r almost always means the log return.

The IID Dream — Why We Chase Clean Returns

Week 1 said models learn patterns from examples; those examples should behave

What models quietly assume

- Rows are 'IID-ish': similar behavior, not wildly dependent.
- Stable variance — not calm Monday, hurricane Tuesday.
- Roughly bell-curved errors (Week 1's noise term).

What dollar bars deliver

- Each row carries a similar amount of market activity.
- Variance per bar becomes far more stable.
- Fat tails shrink toward the bell curve — the model's food gets digestible.

Bars are not cosmetics — they decide whether Week 1's assumptions hold at all.

Time Bars vs. Dollar Bars — What Changes

The statistical scoreboard you will reproduce in Lab 3

Property	Time bars (daily)	Dollar bars
Variance over time	Regime-dependent — calm vs. wild stretches	Far more stable per bar
Tails (kurtosis)	Fat — extreme rows too common	Closer to the bell curve
Busy weeks	Same 5 rows as a quiet week	More rows exactly when info flows
ML consequence	Model over-reacts to regime shifts	Cleaner food → steadier learning

Lab 3 asks Claude to compute this table on real data — you will see the kurtosis drop.

Sampling in Practice — Pros vs. This Course

What a trading desk does, and the honest approximation we make in Colab

On a trading desk

- Raw material: tick data — every trade with price, size, timestamp.
- Bars are built trade by trade (TAQ, exchange feeds).
- Thresholds are tuned per instrument and re-estimated.

In this course

- yfinance gives free daily Close and Volume — no tick feed.
- We build dollar bars from daily dollar volume: same logic, coarser ruler.
- Every concept transfers 1:1 when you later meet real tick data.

Learn the logic on daily data; rent the tick data when the job pays for it.

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PART 2 · EVENT SAMPLING

Focusing on What Matters

The CUSUM event filter

Even With Good Bars, a Problem Remains

Most rows are boring. Why train on them?

Imagine studying for a test by re-reading the blank pages of your textbook. Most market moments are 'blank pages' — nothing important is happening.

Training on everything

- Wastes effort on quiet, meaningless rows.
- Drowns the rare important moves in noise.
- Makes the model worse, not better.

We want a filter that says

- "Wake me up only when something real happens."
- Skip the quiet drift.
- That filter is called CUSUM.

CUSUM = a Smoke Alarm for Prices

Ignore the toast-smoke; sound the alarm on a real fire. CUSUM does this for price moves.

CUSUM Is Like a Smoke Alarm

A smoke alarm ignores normal air

A little smoke from toast? Stays quiet.

Smoke slowly builds up and crosses a danger level? It SOUNDS the alarm.

After you clear the air, it resets and waits for the next real event.

It reacts to meaningful build-up, not to every tiny puff.

CUSUM does the same for prices

Smoke = price drifting in one direction.

Danger level = the threshold h .

Alarm = "record an EVENT here!"

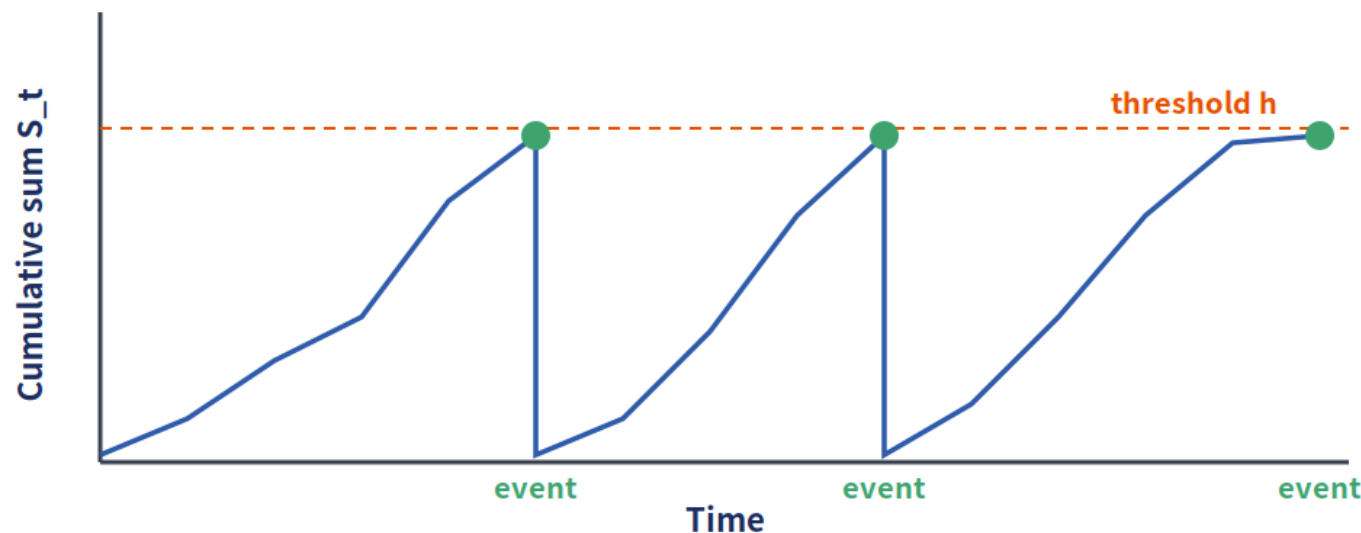
Reset = start counting drift again.

The model studies the alarm moments, not the boring quiet air in between.

How CUSUM Works, Step by Step

A running total that resets on each threshold crossing. Green dots = events the model studies.

The CUSUM Filter: Sample Only When Something Happens



The idea in plain words

1. Keep a running total of how much the price has drifted upward.
2. When the total crosses a threshold, record an EVENT and reset to zero.
3. Ignore the boring sideways drift in between.

Result: the model trains on moments that matter, not on quiet noise.

CUSUM in Two Short Lines

Don't fear it — it just says 'add up the drift; ring when it's big'

$$S_t = \max\{ 0, S_{t-1} + (y_t - \mathbb{E}_{t-1}[y_t]) \}$$

Sample an event when $S_t \geq h$

Reading line 1

- S_t keeps a running total of one-directional drift.
- $\max\{0, \dots\}$ floors it at zero (one-sided).
- Each step adds the newest surprise.

Reading line 2

- When S_t reaches h , ring the alarm: record an event.
- Then reset S_t to 0 and start again.
- Bigger $h \rightarrow$ fewer, only-bigger events.

Choosing the Threshold h

The only knob CUSUM has — think of it as the alarm's sensitivity dial

$h = 1\sigma$ — jumpy

many events

Rings on modest drifts. Lots of training rows, but many are still noise.

use when data is scarce

$h = 2\sigma$ — balanced

the usual start

Fires on moves worth a look, sleeps through chatter. A sensible default.

start here in Lab 2

$h = 3\sigma$ — strict

few, big events

Only genuine shocks pass. Very clean rows — but perhaps too few to learn from.

use for rare-event studies

There is no free lunch: h trades row quantity against row quality — Lab 2 measures the trade.

CUSUM by Hand — Six Days in One Table

$h = 3\%$. Watch the running total build, ring, and reset (full version: this week's problem set)

Day	Return r	$S^+ = \max(0, S^+ + r)$	Alarm? ($S^+ > 3$)
1	+1.0%	1.0	—
2	+2.5%	3.5	EVENT → reset to 0
3	−1.0%	0.0 (floor at zero)	—
4	+2.0%	2.0	—
5	−0.5%	1.5	—
6	+1.0%	2.5	— (still building)

One line of arithmetic per day — that is the whole algorithm. The mirror S^- catches down-moves.

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PART 3 · DIRTY-DATA TRAPS

Dirty-Data Traps

How clean-looking data fakes great backtests

A Strategy That Looks Too Good

If a backtest seems amazing, suspect the data first

It's like bragging you'd ace every past exam — after seeing the answers. Many 'amazing' strategies are secretly peeking at information they couldn't have had.

The danger

- Dirty or peeked-at data inflates results.
- It looks brilliant on paper.
- Then it loses money the moment it goes live.

The discipline

- Know the three classic traps (next slide).
- Check for them before trusting any result.
- Honesty with yourself is a quant's #1 skill.

Three Traps That Fake Great Backtests

Survivorship, look-ahead, and outliers. If it looks too good, suspect the data first.

Three Data Traps That Silently Inflate Your Backtest

1 · Survivorship Bias

Only today's survivors are in your dataset.

Bankrupt or delisted firms vanish from the record.

Effect: your strategy looks great because the losers were erased from history.

Fix: use point-in-time data.

2 · Look-Ahead Bias

Using information that was not yet available.

E.g. using a full-year earnings figure in January, before it was reported.

Effect: impossible profits that vanish in live trading.

Fix: align data to release dates.

3 · Outliers & Gaps

Bad ticks, missing days, unadjusted splits.

A single fat-finger price or a 2-for-1 split can create a fake 50% return.

Fix: use adjusted prices, winsorize extremes, handle gaps explicitly.

Survivorship Bias — the Missing Losers

The most common trap, with a simple picture

Ask today's millionaires how they got rich and you'll hear 'take big risks!' You never hear from the many who took big risks and went broke — they're gone.

In market data

- Datasets often list only firms that still exist today.
- Bankrupt / delisted firms quietly disappear.
- Your strategy looks great — because the losers were erased.

The fix

- Use 'point-in-time' data that includes dead firms.
- Always ask: who is missing from this dataset?
- Survivorship makes the past look easier than it was.

Look-Ahead Bias — Trading on Tomorrow's News

Using information at a time when you could not have had it

Classic slip: compute a signal from today's CLOSE, then pretend you traded at today's close — in real life the number wasn't final until the market shut.

Where it sneaks in

- Same-day close used for a same-day trade decision.
- Earnings numbers as reported today, later restated.
- Index membership lists taken from today, applied to the past.

The fix

- Lag every feature: decide at t , trade at $t+1$.
- Use as-was ('point-in-time') fundamentals.
- Golden question: could I really have known this then?

Outliers — One Weird Row Can Steer the Ship

A single fat-finger print can dominate an entire regression

A stock shows a -80% 'return' — was that a crash, a stock split you forgot to adjust, or a data vendor glitch? The model can't tell; you must.

Why they are dangerous

- Squared-error models (Week 1's MSE) weight big errors heavily.
- One bad print can flip a signal's sign.
- Deleting them blindly can erase real crashes — the rows you care about.

The fix

- Investigate first: split? glitch? genuine crash?
- Adjust prices for splits and dividends (auto_adjust).
- Winsorize / cap only after you understand the cause.

Which Trap Is It? — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Scenario	Which trap?
A backtest on today's S&P 500 members, run over the last 20 years	Survivorship — the 2005 losers are missing
Signal built from today's close, executed at today's close	Look-ahead — the close wasn't known intraday
A -75% daily 'return' that was really a 4-for-1 split	Outlier — unadjusted corporate action
Fund database drops funds that shut down	Survivorship — dead funds erased
Strategy uses final (restated) earnings for past dates	Look-ahead — as-was numbers differed

If you got 4 of 5, the traps can't fool you anymore.

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PART 4 · HANDS ON

Your Turn — Hands On

Build bars and a CUSUM filter in your browser

Getting This Week's Data

Same companion notebook — Week 2 loaders live in Section 3

From the companion notebook (§ 3)

- Open FML_Data_Setup_Guide.ipynb in Colab.
- Run Setup (§ 0), then the § 3 loader cell.
- You get ~1 year of daily prices + volume.

What you get back

- bars_raw: Close, Volume, dollar_vol columns.
- No tick data needed — daily bars are fine for learning.
- Offline backup (§ 5) gives a simulated series.

[CODE] FML_Data_Setup_Guide.ipynb · § 3(a) Week 2 loader

```
import yfinance as yf
bars_raw = yf.download('AAPL', period='1y', auto_adjust=True,
                        multi_level_index=False, progress=False)
bars_raw['dollar_vol'] = bars_raw['Close'] * bars_raw['Volume'] # for dollar bars
```

Hands-On Lab 1 — Build Dollar Bars

Turn raw daily data into activity-based bars

[CODE] colab_lab1.py · data from setup guide § 3

```
import numpy as np, pandas as pd
df = bars_raw.copy()          # from setup guide
df['dv'] = df['Close'] * df['Volume']

thresh = df['dv'].sum() / 50    # aim for ~50 bars
bars, cum, start = [], 0, 0
for i, v in enumerate(df['dv']):
    cum += v
    if cum >= thresh:
        bars.append((df.index[start], df.index[i]))
        cum, start = 0, i + 1

print(f'{len(bars)} dollar bars built')
spans = [(e - s).days for s, e in bars]
print('calendar days per bar:', spans[:10])
```

Your task (step by step)

- Load Close & Volume (§ 3).
- Pick a dollar target (total / 50).
- Walk day by day, summing dollar volume.
- Close a bar each time you hit the target.
- Count calendar days inside each bar.
- Q: are busy stretches split into more bars?

Hands-On Lab 2 — A CUSUM Smoke Alarm

Detect real moves and count how few events there really are

[CODE] colab_lab2.py · data from setup guide § 3

```
import numpy as np
px = bars_raw['Close'].dropna()
ret = np.log(px).diff().dropna()

def cusum_events(r, h):
    s_pos = s_neg = 0; events = []
    for t, x in r.items():
        s_pos = max(0, s_pos + x)
        s_neg = min(0, s_neg + x)
        if s_pos > h: events.append(t); s_pos = 0
        elif s_neg < -h: events.append(t); s_neg = 0
    return events

for k in [1, 2, 3]:
    n = len(cusum_events(ret, k * ret.std()))
    print(f'h = {k} sigma -> {n} events')
```

Your task (step by step)

- Load daily prices (§ 2/ § 3).
- Turn them into log returns.
- Run a symmetric CUSUM with threshold h .
- Try $h = 1x, 2x, 3x$ the return's std.
- Q: how fast does the event count drop as h grows?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 + 2 code)

You are my machine-learning tutor.

- 1) Take my Lab 1 dollar-bar code above and explain every line in one plain-English sentence each.
- 2) Extend it: compute log returns per TIME bar (daily) and per DOLLAR bar, and print a small table comparing their std and excess kurtosis.
- 3) Extend Lab 2: plot event count vs h for $h = 0.5x$ to $3x$ sigma in steps of 0.5.
- 4) In 3 sentences, connect the kurtosis drop to Week 1's bias/variance/noise story.

Do not use any library we have not seen in class.

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do'.

Comprehension Check

Try each question first — the answer is revealed below

Q1 Why are dollar bars often better than time bars?

- A** They are faster to compute
- B** They sample by activity → cleaner returns ✓
- C** They never contain outliers
- D** They need tick data to work

Answer: Dollar bars close by traded value, so returns behave more like the clean bell curve ML expects.

Q2 A backtest only includes firms that still exist today. This is:

- A** Look-ahead bias
- B** Changing variance
- C** Survivorship bias ✓
- D** A dollar bar

Answer: The losers (delisted / bankrupt) are missing, so results look too good — survivorship bias.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 You raise the CUSUM threshold h from 1σ to 3σ . The event count:

- A** Rises — the alarm gets jumpier
- B** Falls — only bigger moves ring ✓
- C** Stays exactly the same
- D** Becomes random

Answer: A higher threshold needs a bigger accumulated drift, so fewer (only larger) events fire.

Q4 A signal uses today's close and trades at today's close. This is:

- A** Survivorship bias
- B** An outlier problem
- C** Look-ahead bias ✓
- D** Perfectly fine

Answer: The close isn't known until the market shuts — deciding with it intraday peeks at the future.

Decide the Bars BEFORE the Labels

Week 2 sits at the very front of the meta-strategy production line

Bars (rows) → Events (which rows matter) → Labels (Week 3) → Features & Models

Why order matters

- Every later step inherits the rows you define today.
- A bad sampling choice cannot be fixed by a clever model.
- This is López de Prado's assembly line from Week 1 — each station does one job well.

What Week 3 will assume

- Clean, activity-sampled bars (today's dollar bars).
- A short list of event timestamps (today's CUSUM).
- On top of these we will define WHAT to predict: the Triple-Barrier label.

Two New Tools, in One Picture Each

Both share one theme: let the DATA decide when something is worth noticing

1

Dollar bars = party photos taken on action

Close a row by traded value, not by the clock. Cleaner, more IID-like returns to feed models.

2

CUSUM = a smoke alarm for prices

Add up drift; ring (record an event) only when it crosses threshold h . Train on what matters.

Keep the two pictures — party photos and a smoke alarm — and Week 2 is yours.

Week 2 — Three Things to Remember

Before next class: finish Labs 1–3 and note how the event count changed with h

1

How you cut the data is a real choice

Time bars are convenient but statistically poor. Dollar bars sample by activity → cleaner returns.

2

Sample events, not every quiet row

CUSUM is a smoke alarm: it fires only on a real move, focusing the model on what matters.

3

Dirty data fakes great backtests

Survivorship, look-ahead, and outliers inflate results. Always suspect the data before the model.

Next week — Labeling & Meta-Labeling: defining WHAT to predict (Triple-Barrier).